Learning from Delayed Semi-Bandit Feedback under Strong Fairness Guarantees

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***Abstract*—Multi-armed bandit frameworks, including combinatorial semi-bandits and sleeping bandits, are commonly employed to model problems in communication networks and other engineering domains. In such problems, feedback to the learning agent is often delayed (e.g. communication delays in a wireless network or conversion delays in online advertising). Moreover, arms in a bandit problem often represent entities required to be treated fairly, i.e. the arms should be played at least a required fraction of the time. In contrast to the previously studied asymptotic fairness, many real-time systems require such fairness guarantees to hold even in the short-term (e.g. ensuring the credibility of information flows in an industrial Internet of Things (IoT) system). To that end, we develop the Learning with Delays under Fairness (LDF) algorithm to solve combinatorial semi-bandit problems with sleeping arms and delayed feedback, which we prove guarantees strong (short-term) fairness. While previous theoretical work on bandit problems with delayed feedback typically derive instance-dependent regret bounds, this approach proves to be challenging when simultaneously considering fairness. We instead derive a novel instance-independent regret bound in this setting which agrees with state-of-the-art bounds. We verify our theoretical results with extensive simulations using both synthetic and real-world datasets.**

I. Introduction

A variety of modern problems in communication networks and other engineering fields involve optimizing some time-varying quantity with unknown statistics. Many such problems are good candidates to be modeled as *multi-armed bandit* (MAB or simply bandit) problems, in which the player plays an action (also called an arm) in each round and receives a random reward from some unknown distribution. Thus, the player must play arms in order to accumulate reward over time (exploitation), while simultaneously learning which arms yield the most reward (exploration).

A practical generalization of the original MAB problem, called *combinatorial semi-bandits* [1], allows for multiple arms to be played in each round and the player to receive the reward from each arm individually (semi-bandit feedback). Here, the player plays a subset of the arms, called a *super-arm*, following some combinatorial structure (e.g. subgraphs of a graph or knapsack packings). For example, consider the problem of link scheduling in wireless networks, in which the successful transmission of a packet over a link yields a reward based on the importance of its contents. In this setting, non-interfering links can be scheduled simultaneously, and the feedback of their transmissions are received individually. Here, arms represent links and super-arms represent a set of links that can be scheduled simultaneously. Additionally, links may not have packets to transmit or may be experiencing poor channel conditions at times, preventing them from being scheduled. This property is captured in another practical extension called *sleeping bandits* [2], in which arms could be randomly unavailable in each round. The combined *combinatorial sleeping bandit* setting can be applied to a wide range of problems from network resource allocation to online advertising [3]. In a typical bandit setup, it is assumed that the player receives the reward from decisions immediately after making them. However, in many real world applications, this feedback is often delayed. For example, a wireless network is subject to communication delays and packets may be transmitted before receiving the feedback from previous transmissions. As another example, consider the problem of deciding which ads to display to a user on a webpage in order to maximize sales. After having clicked an ad for a product, a user intending to purchase it may not do so immediately. Such delays are non-trivial and can range from hours to days [4]. Then, considering *delayed feedback* opens the combinatorial sleeping bandit framework to an even wider range of applications.

In many bandit problems, arms represent users or organizations to be served. Yet another practical modeling consideration is then to guarantee *fairness* to users in a system. For example, we would like to guarantee quality of service (QoS) when scheduling links in a wireless network and we would like to guarantee to advertisers that their ads will be displayed with a certain frequency. To that end, [3] formalizes the notion of fairness in bandit problems as guaranteeing a minimum selection fraction of arms. In particular, an asymptotic constraint is introduced such that each arm is played at least the required fraction of the time in the long run (as time tends to infinity).

However, in many real-world problems, we would like such fairness guarantees to hold even in the short term. For example, in a real-time wireless network, we would like to guarantee QoS while also ruling out large short-term performance fluctuations [5]. Short-term fairness of this sort is particularly important in emerging fields such as industrial IoT and virtual reality (VR), in which packets from the recent past are required to accurately estimate the real-time status of information sources [6]. Short-term fairness of service to

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edge nodes is also a desirable guarantee in edge computing systems [7]. To address the concern of short-term fairness, a stronger notion of fairness, called α-fairness, is introduced in [8], which guarantees that the difference between the required number of plays and the actual number of plays of an arm is no more than α for each and every round.

The *Upper Confidence Bound* (UCB) family of algorithms is known to successfully handle the previously mentioned exploration vs. exploitation dilemma intrinsic to bandit problems. Both [3] and [8] develop UCB-based algorithms that guarantee their respective notions of fairness. In [3], the *Learning with Fairness Guarantee* (LFG) algorithm is developed for the combinatorial sleeping bandit setting, which uses a *virtual queue* to keep track of the cumulative unfairness to each arm. The algorithm works by playing the super-arm with the largest combination of UCB and unfairness, with a tuning parameter to balance the two conflicting goals of reward and fairness. LFG is proven to achieve asymptotic fairness. To guarantee the stronger notion of α-fairness, the *Fair-UCB* algorithm is developed in [8] for the regular MAB setting. This algorithm works by playing the arm with the highest UCB if there is no unfairness to any arm, or playing the arm with the greatest unfairness otherwise. Note that [8] does not consider combinatorial or sleeping arms. While the Fair-UCB guarantees α-fairness for any initially specified value of α, the LFG algorithm is able to simultaneously address both the reward accumulation and fairness goals in each round, instead of switching between the two. Therefore, it is interesting to investigate how to unify these two conflicting goals under α-fairness while taking feedback delays into consideration, which adds another dimension of complexity to the problem.

In this paper, we develop the *Learning with Delays under Fairness* (LDF) algorithm to guarantee α-fairness while maximizing reward in the presence of feedback delays. We rigorously prove that our algorithm guarantees α-fairness, and derive an upper bound on the *regret* of LDF, where regret is defined as the cumulative difference between the player's reward and the reward accrued by a genie that has full knowledge of the reward distributions and plays optimally. While previous regret bounds for bandit problems with delayed feedback depend on the particular problem instance, such an approach proves to be challenging when simultaneously considering fairness. Instead, we employ novel techniques to derive an instance-independent regret bound. Our regret bound is on the same order as state-of-the-art bounds, and the delays only contribute an additive penalty to the regret, which agrees with the observations of [9]. To the best of our knowledge, this work is the first to study fairness and delayed feedback together in a bandit problem. We also verify our theoretical results with extensive simulations both on synthetic data and on a real-world dataset.

The rest of the paper is organized as follows. In Section II, we discuss the related work. In Section III, we outline three timely applications that motivate this work. In Section IV, we formulate the problem of combinatorial sleeping bandits with delayed feedback and define α-fairness accordingly. In Section V, we introduce the LDF algorithm and present our theoretical results. In Section VI, we present our empirical results on synthetic and real-world data and in Section VII, we conclude the paper and discuss future research directions.

II. Related Work

The asymptotic fairness constraint first established in [3] has been recently applied to many communications settings. For example, [10] and [11] apply the framework to fair client selection in a federated learning system, and [12] studies fair worker recruitment in mobile crowdsensing. Closely related constraints regarding long-term time-average resource consumption, where playing an arm incurs some random cost, have also recently been considered in the communications literature. For example, [13] considers the problem of caching files with unknown file popularity subject to caching cost constraints, and [14] considers the problem of optimizing computation delay in mobile edge computing subject to energy consumption constraints. These works employ algorithms designed in the same vein as LFG [3]. The stronger notion of α-fairness introduced by [8] has so far gained little attention in engineering applications. [15] considers linear bandits with general constraints that encompass some notion of fairness.

Delayed feedback has attracted much attention in communications research since it was introduced in the bandit literature by [9]. For example, [16], [17], and [18] study computation offloading in fog networks where the processing time is subject to delays. [19] considers the effect of transmission delays on rate adaptation in wireless networks and [20] studies the problem of task offloading in vehicular networks with task assignment delay. [21] considers the impact of delayed feedback from users on file popularity in the cache placement problem, and [22] studies the impact of delayed feedback on mobility management in wireless networks. While many settings consider bounded delays, theoretical guarantees for arbitrary delay distributions have also been derived [23], [24].

III. Motivating Applications

Guaranteeing α-fairness benefits many of the applications considered in recent work applying the combinatorial sleeping bandits with fairness constraints framework to engineering problems. Moreover, many of those applications could more accurately be modeled by considering delayed feedback. In this section, we present some real-world applications of combinatorial sleeping bandits in which short-term fairness and delayed feedback are important modeling considerations.

A. Conversion Rate Optimization in Online Advertising

The textbook example for a real-world application of bandits is online ad placement using a pay-per-click model, where arms represent ads and a Bernoulli reward represents a successful click, resulting in the advertiser paying the website [25]. However, the *pay-per-conversion* model is preferable to advertisers. In this payment model, the advertiser only pays the website if a click is successfully converted into a sale. Here, a Bernoulli reward represents a successful conversion and is

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subject to a delay ranging from hours to days [4]. Furthermore, we wish to only show the user relevant ads, so only a subset of the ads is available to be shown for each page visit. Each webpage may also have several areas in which ads can be placed, so the ad placement system must choose a combination of the available ads according to the page layout. Finally, α-fairness is a marketable guarantee to advertisers, who would like that their ads always be shown with a certain frequency.

B. Caching by User Preference in HetNets

Building on the setting in [21], consider a heterogeneous cellular network consisting of a macro-cell base station (MBS) and several small-cell base stations (SBSs) able to store cached files. In each time slot, the MBS must decide a set of files to cache, called a *caching policy*, which depends on the size of the files and the storage capacity of the SBSs, as well as the users' availability to consume the files. Here, an arm represents a file, a super-arm represents a caching policy, and the availability of arms depends on the availability of users. Reward represents the aggregated ratings or other engagement metrics given by the users requesting the file, which is subject to delays since the users must first finish consuming the file. Additionally, one would like to cache files with some fairness, so that even less-preferred files may be promptly delivered to the users requesting them. α-fairness is also preferable here to asymptotic fairness to minimize short-term delays.

C. Task Assignment in Crowdsourcing

Building on one of the applications presented in [3], consider a crowdsourcing platform in which tasks arrive and must be assigned to one or many workers. Here, a super-arm represents a group of workers, each of whom may be unavailable to accept work when the task arrives. Reward represents the quality of work done on the task by each worker, which is subject to delays since each task takes some amount of time to complete. Additionally, we would like to guarantee some short-term fairness of tasks allocated to each of the workers, so that they maintain interest in the platform.

IV. Problem Formulation

We consider a stochastic combinatorial semi-bandit problem with sleeping arms and delayed semi-bandit feedback. A *stochastic combinatorial semi-bandit* problem [1] consists of a set of arms $\mathcal{N} \triangleq \{1, 2, \ldots, N\}$ and a set of combinations of those arms $\mathbb{S} \subseteq 2^{\mathcal{N}}$, where $2^{(\cdot)}$ denotes the power-set and each $\mathcal{S} \in \mathbb{S}$ is called a *super-arm*. We assume $|\mathcal{S}| \leq M$ for all $\mathcal{S} \in \mathbb{S}$, where M represents the size of the largest super-arm. [1] Each arm $i \in \mathcal{N}$ is associated with a sequence of i.i.d. random variables $X_i(1), X_i(2), \ldots$ each with support $[0, 1]$ and unknown a priori mean μ_i, where $X_i(t)$ represents the reward for playing arm i in round t. The term "semi-bandit" refers to the fact that the player receives feedback from each arm individually, rather than the aggregated feedback of the super-arm played.

We also consider *sleeping arms* such that at the start of each round t, a set of available arms $\mathcal{A}(t) \subseteq \mathcal{N}$ is randomly drawn and revealed to the player. Then, the player is restricted to playing a super-arm that only contains available arms, i.e. super-arms belonging to the set $\mathbb{S}_{\mathcal{A}(t)} \triangleq 2^{\mathcal{A}(t)} \cap \mathbb{S}$. We let $\mathcal{S}(t) \in \mathbb{S}_{\mathcal{A}(t)}$ denote the super-arm chosen by the player in round t and we assume that $\mathcal{A}(t)$ is stationary with probability density function $p(\mathcal{A}) \triangleq \mathrm{P}(\mathcal{A}(t) = \mathcal{A})$.

Feedback is assumed to be delayed such that when arm i is played in round t, the player actually receives the reward $X_i(t)$ in round $t + \delta_i(t)$ where $\delta_i(t) \in \{0, 1, 2, \ldots\}$ is a random delay. We do not require that the delays are i.i.d. over time. In fact, the only assumption about the delays is that their distributions are light-tailed, i.e. there exist constants $0 < \theta, D < \infty$ such that $\mathbf{E}\left[e^{\theta \delta_i(t)}\right] \leq D$ for all arms i and rounds t. This assumption is valid in many applications. For example, the delay between a click and a sale in the online advertising setting discussed in Section III is found to be exponentially distributed [4]. Then, at the end of each round t, the player receives the vector of rewards from previous rounds $(X_i(\tau))_{\tau \leq t : i \in \mathcal{S}(\tau), \tau + \delta_i(\tau) = t}$ from each arm i. The primary goal of the player is to maximize the cumulative reward $\sum_{t=1}^{T} \sum_{i \in \mathcal{S}(t)} X_i(t)$ up to a *time horizon* T.

In addition to the goal of reward accumulation, we also consider the goal of guaranteeing *fairness*, i.e., guaranteeing that each arm is played at least a required fraction of the previous rounds. Let r_i denote the fairness requirement of arm i and let $n_i(t) \triangleq \sum_{\tau=1}^{t} d_i(\tau)$ denote the number of times arm i was played up to and including round t, where $d_i(t) \triangleq \mathbb{1}_{\{i \in \mathcal{S}(t)\}}$ and $\mathbb{1}_{\{\cdot\}}$ denotes the indicator variable of an event. We say an algorithm is α-fair for some $\alpha \geq 0$ if

$$r_i \, t - \mathbf{E}[n_i(t)] \leq \alpha \quad \forall \, i \in \mathcal{N}, \, t \in \{1, 2, \ldots\}, \tag{1}$$

where α represents the *unfairness*. Note that this constraint is *strong* in the sense that it guarantees fairness in each round, unlike the asymptotic guarantee in [3]. [2] If $\alpha = 0$, then the arms are expected to be played completely fairly according to the requirements $(r_i)_{i \in \mathcal{N}}$. We note that our definition of α-fairness differs from [8] in that their definition holds almost surely, while ours only holds in expectation. The reason for this difference is that their definition of α-fairness would be impossible to guarantee with sleeping arms, since an arm could be randomly unavailable for a long enough sequence of rounds such that the constraint is violated.

As done in [3], we evaluate an algorithm in the goal of reward accumulation by comparing it to a completely fair genie having full knowledge of the reward distributions that plays according to an optimal randomized policy. In particular, the genie observes the set of available arms $\mathcal{A}(t)$ in each round

[1] If we let $\mathbb{S} = \left\{\mathcal{S} \in 2^{\mathcal{N}} : |\mathcal{S}| \leq M\right\}$, then we recover the set of super-arms considered by [3]. However, our setting allows for an arbitrary set of super-arms, as in the usual combinatorial semi-bandit setup.

[2] [3] considered the constraint that $\liminf_{t \to \infty} \frac{\mathbf{E}[n_i(t)]}{t} \geq r_i$. Clearly, this is satisfied when $r_i \, t - \mathbf{E}[n_i(t)] \leq f(t)$ for some function f having $\lim_{t \to \infty} \frac{f(t)}{t} = 0$, with a constant function $f(t) = \alpha$ giving a particularly strong guarantee (α-fairness).

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t and randomly plays a super-arm $\mathcal{S} \in \mathbb{S}_{\mathcal{A}(t)}$ with probability $q^*(\mathcal{S} \mid \mathcal{A}(t))$, where the distribution q^* is the optimal solution to the following problem:

$$\begin{aligned} \max_{q} \quad & \sum_{\mathcal{A}\in 2^{\mathcal{N}}} p(\mathcal{A}) \sum_{\mathcal{S}\in\mathbb{S}_{\mathcal{A}}} q(\mathcal{S}\mid\mathcal{A}) \sum_{i\in\mathcal{S}} \mu_i \\ \text{s.t.} \quad & \sum_{\mathcal{A}\in 2^{\mathcal{N}}} p(\mathcal{A}) \sum_{\mathcal{S}\in\mathbb{S}_{\mathcal{A}}: i\in\mathcal{S}} q(\mathcal{S}\mid\mathcal{A}) \geq r_i, \qquad i \in \mathcal{N} \\ & q(\,\cdot\mid\mathcal{A}) \text{ is a categorical distribution on } \mathbb{S}_{\mathcal{A}}, \quad \mathcal{A}\in 2^{\mathcal{N}}. \end{aligned} \tag{2}$$

We say the fairness requirement vector $(r_i)_{i\in\mathcal{N}}$ is *supportable* if (2) has a feasible solution. Throughout the rest of this paper, we assume that there exists $\varepsilon > 0$ so that the fairness requirement vector $(r_i + \varepsilon)_{i\in\mathcal{N}}$ is also supportable.

Let $\mathcal{S}^*(t)$ denote the super-arm played by the genie in round t and let $d_i^*(t) \triangleq \mathbb{1}_{\{i\in\mathcal{S}_i^*(t)\}}$ indicate that the genie plays arm i in round t. Then the *regret* of the player employing a bandit algorithm ALG is given by

$$\mathrm{R}^{\mathrm{ALG}}(T) \triangleq \sum_{t=1}^{T} \sum_{i\in\mathcal{N}} \mu_i \, \mathbf{E}[d_i^*(t) - d_i(t)]. \tag{3}$$

To summarize, regret measures the expected difference in reward between the genie, who plays a completely fair ($\alpha = 0$ according to (1) and (2)) optimal policy, and the player. We therefore expect that playing a less fair policy that places more emphasis on reward will result in lower regret.

V. Algorithm Design and Theoretical Results

While the *Fair-UCB* [8] algorithm guarantees α-fairness for any user-defined value of α in the stochastic multi-armed bandit setting, we decide against using it because it can result in very poor performance in the extended combinatorial sleeping bandit setting. We demonstrate this through simulation in Section VI. Instead, we develop an algorithm following the design philosophy of works such as [3] and [26] which combines UCB with a virtual queue to track unfairness.

A. The Learning with Delays under Fairness Algorithm

Let $s_i(t) \triangleq \sum_{\tau\leq t:\tau+\delta_i(\tau)\leq t} d_i(\tau)$ denote the number of rewards received from arm i up to and including round t. Recall $n_i(t) = \sum_{\tau=1}^{t} d_i(\tau)$ is the number of times arm i has been played up to and including round t, so we always have $s_i(t) \leq n_i(t)$. We let $g_i(t) \triangleq n_i(t) - s_i(t)$ denote the number of outstanding rewards from arm i in round t. We present the *Learning with Delays under Fairness* (LDF) algorithm below, which is conceptually similar to the LFG algorithm [3] but with modifications to accommodate delayed feedback.

In each round t, after observing the set of available arms $\mathcal{A}(t)$, the player employing LDF plays the super-arm

$$\mathcal{S}(t) \in \operatorname*{argmax}_{\mathcal{S}\in\mathbb{S}_{\mathcal{A}(t)}} \sum_{i\in\mathcal{S}} (Q_i(t) + \eta U_i(t)), \tag{4}$$

where $Q_i(t)$ is a virtual queue keeping track of the unfairness to arm i in round t, $U_i(t)$ is the UCB estimate of the reward from arm i in round t, and $\eta > 0$ is a tuning parameter to balance the tradeoff between reward and fairness.

Each component of the vector of virtual queues $\mathbf{Q}(t) \triangleq (Q_i(t))_{i\in\mathcal{N}}$ is initialized by $Q_i(1) = 0$ and is updated at the end of each round t by $Q_i(t+1) = [Q_i(t) + r_i - d_i(t)]^+$ where $[\,\cdot\,]^+$ denotes $\max(\,\cdot\,, 0)$.

Remark 1. Expanding the virtual queue update gives

$$\begin{aligned} Q_i(t+1) &\geq Q_i(t) + r_i - d_i(t) \\ &\geq Q_i(t-1) + 2r_i - \textstyle\sum_{\tau=t-1}^{t} d_i(\tau) \\ &\geq Q_i(t-2) + 3r_i - \textstyle\sum_{\tau=t-2}^{t} d_i(\tau) \\ &\;\;\vdots \\ &\geq Q_i(1) + r_i\, t - \textstyle\sum_{\tau=1}^{t} d_i(\tau) = r_i\, t - n_i(t). \end{aligned} \tag{5}$$

Then, LDF is α-fair if $\mathbf{E}[\mathbf{Q}(t+1)] \preceq \alpha$ for all rounds t.

Let $\hat{\mu}_i(t)$ denote the empirical mean of the $s_i(t)$ rewards received from arm i up to and including round t, which is defined whenever $s_i(t) > 0$. Then the UCB of arm i at the start of round t is defined as

$$U_i(t) \triangleq \begin{cases} \min\Big(\hat{\mu}_i(t-1) + \sqrt{\frac{3\log t}{2 s_i(t-1)}}, 1\Big) & s_i(t-1) > 0 \\ 1 & s_i(t-1) = 0 \end{cases} \tag{6}$$

where $\sqrt{\frac{3\log t}{2 s_i(t-1)}}$ is the exploration bonus and we clip $U_i(t)$ to 1 since the rewards have $[0, 1]$ support. We initialize $U_i(t) = 1$ when no rewards have yet been received in order to encourage early exploration of all arms.

B. Theoretical Guarantees for the LDF Algorithm

In this section, we first show that LDF is α^{LDF}-fair. We then derive an instance-independent bound on the regret $\mathrm{R}^{\mathrm{LDF}}(T)$ that is, to our knowledge, the first instance-independent bound for a stochastic combinatorial bandit problem with arbitrary light-tailed delays.

Theorem 1. *LDF is α^{LDF}-fair for*

$$\alpha^{\mathrm{LDF}} = \frac{N + 2\eta M}{\varepsilon} - \frac{\sqrt{N}\log\Big(\log\Big(\frac{\varepsilon+2\sqrt{N}}{2\sqrt{N}}\Big) - \frac{\varepsilon}{\varepsilon+2\sqrt{N}}\Big)}{\log\Big(\frac{\varepsilon+2\sqrt{N}}{2\sqrt{N}}\Big)}$$

where $\varepsilon > 0$ is taken to minimize the above and satisfies the condition that $(r_i + \varepsilon)_{i\in\mathcal{N}}$ is a supportable fairness requirement vector.

Proof. We derive drift conditions on the Euclidean norm of the virtual queues $\|\mathbf{Q}(t)\|$. In particular, that its drift $\|\mathbf{Q}(t+1)\| - \|\mathbf{Q}(t)\|$ is bounded by a constant and is expected to become negative when $\|\mathbf{Q}(t)\|$ is large enough. Then, we use techniques developed by [27] to bound the moment-generating function of a process with such drift conditions. Finally, from Jensen's inequality and Remark 1, we obtain α^{LDF}. A detailed proof can be found in Appendix A. □

Notice that α^{LDF} is a linear function of the tuning parameter η, with a slope on the order of M/ε and a vertical intercept given in terms of N and ε. Then we expect to incur a linear increase in unfairness when tuning η to favor reward. We also expect greater unfairness for a greater number of arms N or

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when lesser fairness requirements can be supported (smaller ε). However, an increase in M does not necessarily lead to an increase in unfairness because if more arms can be played in a super-arm (larger M), then greater fairness requirements may be supported (larger ε). So, the factor M/ε does not necessarily increase when M increases and depends heavily on the fairness requirements, the combinatorial structure of the super-arms, and their availability distributions. We note that Theorem 1 does not capture the impact of delay on the unfairness, but show empirically in Section VI that delay only contributes a small additive penalty. It could be very difficult to derive a tight enough α^{LDF} to capture this impact, which remains an open problem.

Theorem 2. *Recall that* $\mathbf{E}\big[e^{\theta\delta_i(t)}\big] \leq D$ *for all arms* i *and rounds* t*. If* T *is small enough that* $T/\log T \leq N/(M\theta^2)$*, then* $\mathrm{R}^{\mathrm{LDF}}(T) \leq N\log T/\theta^2$*. Otherwise,*

$$\mathrm{R}^{\mathrm{LDF}}(T) \leq \frac{NT}{2\eta} + ND + \sqrt{48NMT\log T} + \frac{3+\pi^2}{3}N.$$

Proof. The proof proceeds similarly to those for similar algorithms (see [3] and [26]) with careful modifications to account for the delayed feedback. In order to deal with the delays, we consider the events that all the delays are bounded by $\sqrt{4MT\log T/N}$ and we leverage the light-tailed assumption to deal with the complement events by applying the Chernoff bound. A detailed proof can be found in Appendix B. □

The first term in the bound given in Theorem 2 captures the contribution of fairness, the second term captures the contribution of delayed feedback, and the remaining terms capture the contribution of UCB learning. The impact of delay on the regret is additive, which agrees with the instance-dependent bound given by [9], albeit in terms of the moment-generating function instead of the mean. Moreover, if $\eta \in \Omega\Big(\sqrt{NT/(M\log T)}\Big)$, then $\mathrm{R}^{\mathrm{LDF}}(T) \in \mathcal{O}\big(\sqrt{NMT\log T}\big)$, which is the typical instance independent bound for combinatorial semi-bandits [1].

Although knowledge of a time horizon T is not required to run LDF, Theorems 1 and 2 show how η can be tuned in terms of T to balance regret and fairness. For example, both will be on the order $\mathcal{O}(\sqrt{T})$ if we set $\eta = \sqrt{T}$.

VI. Empirical Results

In this section, we conduct experiments with synthetic data to verify the theoretical properties of the LDF algorithm and we simulate the online advertising setting from Section III using Criteo's conversion log dataset. All experiments are repeated several times and their means are shown along with their 95% confidence intervals (red error bars).

A. Experiments with Synthetic Data

For our synthetic experiments, we use the same problem parameters as in [3], namely $\mathbb{S} = \{\{1,2\},\{2,3\},\{1,3\}\}$, $p(\mathcal{A}) = \prod_{i\in\mathcal{A}} a_i \prod_{i\notin\mathcal{A}}(1-a_i)$ where $\mathbf{a} = (0.9, 0.8, 0.7)$, $\mu = (0.4, 0.5, 0.7)$ and $\mathbf{r} = (0.5, 0.6, 0.4)$. We assume i.i.d. over time geometrically distributed delays and we vary the vector of mean delays $\bar{\delta}$ to observe the impact of delays on the unfairness and regret of LDF. We set time horizon $T = 5000$.

Fig. 1a shows the unfairness for increasing values of η and different delay distributions, where the unfairness is calculated as $\max_{i\in\mathcal{N}, t\in\{1,\ldots,T\}}[r_i\, t - n_i(t)]^+$. We also show the lines of best fit for these curves. These experiments are repeated 1000 times. If the value for α^{LDF} given in Theorem 1 is reasonable, we should observe a linear relationship between η and the unfairness, which we do in fact observe. Also notice that the impact of delays is additive.

Next, we record $\mathrm{R}^{\mathrm{LDF}}(t)$ and $\|\mathbf{Q}(t)\|$ (recall Remark 1) over time for different delay distributions and values of η. These experiments are repeated 10000 times. In Fig. 1b, we observe that the delays vertically shift the regret curve, indicating an additive contribution to the regret, and we notice that the value of η actually changes the shape of the curve. This agrees with our theoretical regret bound (Theorem 2). In Fig. 1c, we notice that while the delays can have an effect on the queue lengths in the short-term, they appear to converge toward a similar value in the long run, and that again, η determines the value that the queue lengths converge to.

B. Experiments with Criteo's Conversion Log Dataset

We simulate the online advertising setting using the conversion log dataset [28] collected by the advertising company Criteo. Each row corresponds to an ad click after a user is shown an ad for a particular product and indicates whether or not the click was converted into a sale, as well as the time delay of the conversion.

We consider a webpage with $M = 2$ ad slots on which $N = 3$ companies wish to display their ads. We take the products with the most data available for them in the dataset and uniformly sample rows corresponding to these products for each click. Since the click rate of the products are not provided, we assume each product has a click rate of 5%. The dataset provides the number of clicks in the last week, which we use along with the assumed click rate to calculate the time between page visits. We assume each product is relevant to a visiting user 85% of the time and we consider two scenarios (i) each product should be shown at least 50% of the time, and (ii) each product should be shown at least 60% of the time.

We compare the LDF algorithm against the Fair-UCB algorithm [8] modified for the more general setting we consider, which plays the super-arm

$$\mathcal{S}(t) \in \begin{cases} \operatorname*{argmax}_{\mathcal{S}\in\mathbb{S}_{\mathcal{A}(t)}} \sum_{i\in\mathcal{S}} U_i(t) & \text{if for all } i \in \mathcal{A}(t), \\ & r_i\, t - n_i(t) \leq \eta \\ \operatorname*{argmax}_{\mathcal{S}\in\mathbb{S}_{\mathcal{A}(t)}} \sum_{i\in\mathcal{S}} [r_i\, t - n_i(t)] & \text{otherwise.} \end{cases} \tag{7}$$

We run the simulations for $T = 5\times 10^5$ impressions (page visits) and set the tuning parameter $\eta = \sqrt{T}$. These experiments are repeated 100 times. We can see from Fig. 1d that Fair-UCB outperforms LDF when the fairness requirements are small, although both algorithms appear to achieve sublinear

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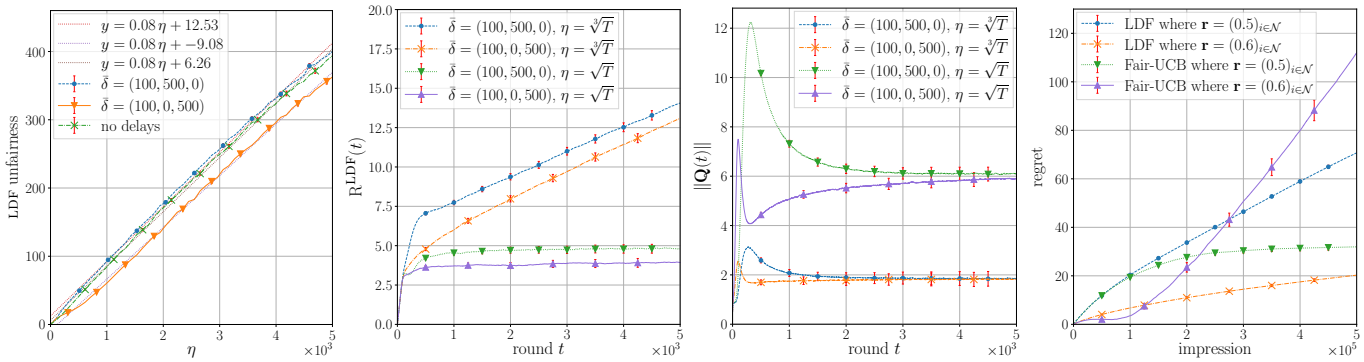

(a) Synthetic data: unfairness vs. η (b) Synthetic data: $\mathrm{R}^{\mathrm{LDF}}(t)$ vs. t (c) Synthetic data: $\|\mathbf{Q}(t)\|$ vs. t (d) Criteo dataset: regret vs. impression

Fig. 1. Experiments evaluating LDF with synthetic data (a), (b), (c), and regret in the online advertising simulations comparing LDF to Fair-UCB (d).

regret. However, when the fairness requirements are larger, Fair-UCB appears to suffer linear regret. We infer from this that Fair-UCB does not extend nicely to the combinatorial sleeping bandit with delayed feedback setting. Table I shows the unfairness under the various simulation parameters. Again, both algorithms perform well, achieving unfairness much less than η when the fairness requirements are small. When the fairness requirements are larger, LDF still achieves unfairness much less than η, while Fair-UCB achieves unfairness approximately η as it was designed to [8]. This observation for the unfairness of LDF is consistent with the results from our experiments with synthetic data.

TABLE I
UNFAIRNESS IN THE CRITEO DATASET SIMULATIONS

Algorithm	Requirements $\mathbf{r}$	Unfairness
LDF	$(0.5)_{i\in\mathcal{N}}$	$0.00024\,\eta \pm 0.05916$
LDF	$(0.6)_{i\in\mathcal{N}}$	$0.00079\,\eta \pm 0.10327$
Fair-UCB	$(0.5)_{i\in\mathcal{N}}$	$0.00069\,\eta \pm 0.08541$
Fair-UCB	$(0.6)_{i\in\mathcal{N}}$	$1.00556\,\eta \pm 0.09092$

VII. Concluding Remarks and Future Work

In this paper, we developed the LDF algorithm and proved that it is α-fair and achieves low regret in the combinatorial semi-bandits with sleeping arms and delayed feedback setting. Our theoretical results are motivated by several timely applications and are supported by extensive simulations.

In future work, we would like to extend the delayed feedback model to consider the possibility of rewards being permanently lost [29]. This extension is relevant to online advertising, in which a user may never follow through on his initial interest in a product, and and to wireless networks, in which packets may be permanently lost. Another possible extension is to relax the stationary availability assumption for sleeping arms and consider the case when availability depends on the delay of the previous play, which models scenarios where arms represent resources that are busy for the duration of the delay. Finally, it would be valuable to consider more general constraints as in [15].

Appendix

A. Proof of Theorem 1

Proof. Fix round t and $\varepsilon > 0$ such that $(r_i + \varepsilon)_{i\in\mathcal{N}}$ is supportable. Define the Lyapunov function $V(t) \triangleq \frac{1}{2}\sum_{i\in\mathcal{N}} Q_i^2(t)$ and its drift $\Delta V(t) \triangleq V(t+1) - V(t)$. Then we have

$$\begin{aligned}\Delta V(t) &\le \frac{1}{2}\sum_{i\in\mathcal{N}}(Q_i(t) + r_i - d_i(t))^2 - \frac{1}{2}\sum_{i\in\mathcal{N}} Q_i^2(t)\\ &= \sum_{i\in\mathcal{N}} Q_i(t)(r_i - d_i(t)) + \sum_{i\in\mathcal{N}}\frac{(r_i - d_i(t))^2}{2}\\ &\le \sum_{i\in\mathcal{N}} Q_i(t)(r_i - d_i(t)) + \frac{N}{2}.\end{aligned} \tag{8}$$

Taking expectations conditioned on $\mathbf{Q}(t)$ gives

$$\begin{aligned}\mathbf{E}[\Delta V(t) \mid \mathbf{Q}(t)] &\le \mathbf{E}\left[\sum_{i\in\mathcal{N}} Q_i(t)(r_i - d_i(t)) + \frac{N}{2} \mid \mathbf{Q}(t)\right]\\ &\overset{(a)}{\le} \frac{N}{2} + \eta M + \sum_{i\in\mathcal{N}} r_i Q_i(t)\\ &\quad - \mathbf{E}\left[\sum_{i\in\mathcal{N}} d_i(t)\left(Q_i(t) + \eta U_i(t)\right) \mid \mathbf{Q}(t)\right]\end{aligned} \tag{9}$$

where in (a), we add the positive quantity $\eta\left(M - \mathbf{E}\left[\sum_{i\in\mathcal{N}} d_i(t)U_i(t) \mid \mathbf{Q}(t)\right]\right)$ and rearrange.

Consider a feasible solution q to (2) for the fairness requirement vector $(r_i+\varepsilon)_{i\in\mathcal{N}}$, which exists by assumption. Let $d_i^q(t)$ indicate that a randomized policy characterized by q plays arm i in round t. Then we can lower bound the conditional expectation on the right-hand side of (9) by

$$\begin{aligned}&\mathbf{E}\left[\sum_{i\in\mathcal{N}} d_i(t)\left(Q_i(t) + \eta U_i(t)\right) \mid \mathbf{Q}(t)\right]\\ &\overset{(a)}{\ge} \mathbf{E}\left[\sum_{i\in\mathcal{N}} d_i^q(t)\left(Q_i(t) + \eta U_i(t)\right) \mid \mathbf{Q}(t)\right]\\ &\ge \mathbf{E}\left[\sum_{i\in\mathcal{N}} d_i^q(t) Q_i(t) \mid \mathbf{Q}(t)\right] \overset{(b)}{\ge} \sum_{i\in\mathcal{N}}(r_i + \varepsilon)Q_i(t)\end{aligned} \tag{10}$$

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where (a) is by definition of the LDF algorithm (4) and (b) is because each $d_i^q(t)$ is independent of $\mathbf{Q}(t)$ and $\mathbf{E}[d_i^q(t)] \geq r_i + \varepsilon$ by the fairness constraint in (2). Then using (10) to upper bound (9) gives

$$\mathbf{E}[\Delta V(t) \mid \mathbf{Q}(t)] \leq \frac{N}{2} + \eta M - \varepsilon \sum_{i \in \mathcal{N}} Q_i(t). \tag{11}$$

Define the drift of the Euclidean norm of the queues $\Delta\|\mathbf{Q}(t)\| \triangleq \|\mathbf{Q}(t+1)\| - \|\mathbf{Q}(t)\|$. Since $\sqrt{\cdot}$ is concave and differentiable, it is upper bounded by its first-order Taylor approximation, meaning that $\sqrt{x_1} - \sqrt{x_2} \leq (x_1 - x_2)\frac{\partial\sqrt{x_2}}{\partial x_2} = \frac{x_1 - x_2}{2\sqrt{x_2}}$ for all $x_1, x_2 > 0$. Then provided $\|\mathbf{Q}(t)\| > 0$, we have

$$\Delta\|\mathbf{Q}(t)\| = \sqrt{2\,V(t+1)} - \sqrt{2\,V(t)} \leq \frac{\Delta V(t)}{\|\mathbf{Q}(t)\|}. \tag{12}$$

Taking expectations conditioned on $\mathbf{Q}(t)$ gives

$$\mathbf{E}[\Delta\|\mathbf{Q}(t)\| \mid \mathbf{Q}(t)] \leq \frac{\mathbf{E}[\Delta V(t) \mid \mathbf{Q}(t)]}{\|\mathbf{Q}(t)\|} \overset{(a)}{\leq} \frac{\frac{N}{2} + \eta M}{\|\mathbf{Q}(t)\|} - \varepsilon \tag{13}$$

where (a) is due to (11) and the fact that $\sum_{i \in \mathcal{N}} Q_i(t) \geq \|\mathbf{Q}(t)\|$. From here we can see that

$$\mathbf{E}[\Delta\|\mathbf{Q}(t)\| \mid \mathbf{Q}(t)]\, \mathbb{1}_{\{\|\mathbf{Q}(t)\| \geq \gamma\}} \leq -\frac{\varepsilon}{2}\, \mathbb{1}_{\{\|\mathbf{Q}(t)\| \geq \gamma\}} \tag{14}$$

where $\gamma \triangleq \frac{2}{\varepsilon}\left(\frac{N}{2} + \eta M\right)$.

For any $\lambda > 0$, we can decompose the moment-generating function of $\|\mathbf{Q}(t+1)\|$ as

$$\mathbf{E}\Big[e^{\lambda\|\mathbf{Q}(t+1)\|}\Big] = \mathbf{E}\Big[C_1 + C_2\, e^{\lambda\|\mathbf{Q}(t)\|}\Big] \tag{15}$$

where $C_1 \triangleq e^{\lambda\|\mathbf{Q}(t)\|}\mathbb{1}_{\{\|\mathbf{Q}(t)\| \leq \gamma\}}\, \mathbf{E}\big[e^{\lambda\Delta\|\mathbf{Q}(t)\|} \mid \mathbf{Q}(t)\big]$ and $C_2 \triangleq \mathbf{E}\big[e^{\lambda\Delta\|\mathbf{Q}(t)\|}\mathbb{1}_{\{\|\mathbf{Q}(t)\| > \gamma\}} \mid \mathbf{Q}(t)\big]$. We bound C_1 by

$$\begin{aligned} C_1 &\leq e^{\lambda\gamma}\, \mathbf{E}\Big[e^{\lambda|\|\mathbf{Q}(t+1)\| - \|\mathbf{Q}(t)\||} \mid \mathbf{Q}(t)\Big] \\ &\overset{(a)}{\leq} e^{\lambda\gamma}\, \mathbf{E}\Big[e^{\lambda\|\mathbf{Q}(t+1) - \mathbf{Q}(t)\|} \mid \mathbf{Q}(t)\Big] \overset{(b)}{\leq} e^{\lambda(\gamma + \sqrt{N})} \end{aligned} \tag{16}$$

where (a) is by the reverse triangle inequality and (b) is because each $(Q_i(t+1) - Q_i(t))^2 \leq 1$. We bound C_2 by

$$\begin{aligned} C_2 &\leq \mathbf{E}\Big[e^{\lambda\Delta\|\mathbf{Q}(t)\|\mathbb{1}_{\{\|\mathbf{Q}(t)\| > \gamma\}}} \mid \mathbf{Q}(t)\Big] \\ &= \mathbf{E}\left[\sum_{x=0}^{\infty} \frac{\big(\lambda\Delta\|\mathbf{Q}(t)\|\mathbb{1}_{\{\|\mathbf{Q}(t)\| > \gamma\}}\big)^x}{x!} \mid \mathbf{Q}(t)\right] \\ &\overset{(a)}{\leq} 1 - \frac{\lambda\varepsilon}{2}\mathbb{1}_{\{\|\mathbf{Q}(t)\| > \gamma\}} + \mathbf{E}\left[\sum_{x=2}^{\infty} \frac{(\lambda|\,\Delta\|\mathbf{Q}(t)\|\,|)^x}{x!} \mid \mathbf{Q}(t)\right] \\ &\overset{(b)}{\leq} 1 - \frac{\lambda\varepsilon}{2}\mathbb{1}_{\{\|\mathbf{Q}(t)\| > \gamma\}} + \left(e^{\lambda\sqrt{N}} - \lambda\sqrt{N} - 1\right) \\ &= e^{\lambda\sqrt{N}} - \lambda\left(\frac{\varepsilon}{2}\mathbb{1}_{\{\|\mathbf{Q}(t)\| > \gamma\}} + \sqrt{N}\right) \end{aligned} \tag{17}$$

where (a) uses (14) to bound the $x = 1$ term and (b) uses the reverse triangle inequality to bound the remaining $x \geq 2$ terms. Define the function $f(\lambda) \triangleq e^{\lambda\sqrt{N}} - \lambda\left(\frac{\varepsilon}{2} + \sqrt{N}\right)$. Notice from (17) that $C_2 \leq f(\lambda)$ in the event $\{\|\mathbf{Q}(t)\| > \gamma\}$ and $C_2 = 0 \leq f(\lambda)$ in the event $\{\|\mathbf{Q}(t)\| \leq \gamma\}$, so in any event, $C_2 \leq f(\lambda)$. Then using the bounds (16) and (17) in (15) gives

$$\begin{aligned} \mathbf{E}\Big[e^{\lambda\|\mathbf{Q}(t+1)\|}\Big] &\leq e^{\lambda(\gamma + \sqrt{N})} + f(\lambda)\, \mathbf{E}\Big[e^{\lambda\|\mathbf{Q}(t)\|}\Big] \\ &\leq e^{\lambda(\gamma + \sqrt{N})} \sum_{\tau=1}^{t} f^{t-\tau}(\lambda) + f^t(\lambda) \\ &\leq e^{\lambda(\gamma + \sqrt{N})} \sum_{\tau=0}^{\infty} f^{\tau}(\lambda). \end{aligned} \tag{18}$$

Taking $\lambda^* \triangleq \operatorname{argmin}_{\lambda > 0} f(\lambda) = \frac{1}{\sqrt{N}} \log\left(\frac{\varepsilon/2 + \sqrt{N}}{\sqrt{N}}\right)$ gives

$$\begin{aligned} &f(\lambda^*) = \frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}} - \frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}} \log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\right) \overset{(a)}{\in} (0, 1) \\ &\implies \mathbf{E}\Big[e^{\lambda^*\|\mathbf{Q}(t+1)\|}\Big] \overset{(b)}{\leq} \frac{e^{\lambda^*(\gamma + \sqrt{N})}}{1 - f(\lambda^*)} \end{aligned} \tag{19}$$

where (a) is because $\sqrt{N} > \varepsilon/2$ and (b) is by taking the closed form of the geometric series. Then by Jensen's inequality, we have

$$\begin{aligned} &\mathbf{E}[\|\mathbf{Q}(t+1)\|] \leq \gamma + \sqrt{N} - \frac{\log(1 - f(\lambda^*))}{\lambda^*} \\ &= \gamma + \sqrt{N} - \frac{\sqrt{N}\log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\right) - \frac{\frac{\varepsilon}{2}}{\sqrt{N}}\right)}{\log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\right)} \\ &= \gamma + \sqrt{N} - \frac{\sqrt{N}\log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\left[\log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\right) - \frac{\frac{\varepsilon}{2}}{\frac{\varepsilon}{2} + \sqrt{N}}\right]\right)}{\log\left(\frac{\frac{\varepsilon}{2} + \sqrt{N}}{\sqrt{N}}\right)} \\ &= \gamma - \frac{\sqrt{N}\log\left(\log\left(\frac{\varepsilon/2 + \sqrt{N}}{\sqrt{N}}\right) - \frac{\varepsilon/2}{\varepsilon/2 + \sqrt{N}}\right)}{\log\left(\frac{\varepsilon/2 + \sqrt{N}}{\sqrt{N}}\right)}. \end{aligned} \tag{20}$$

Substituting $\gamma = \frac{N + 2\eta M}{\varepsilon}$ and recalling Remark 1 finishes the proof. □

B. Proof of Theorem 2

Proof. Notice that we always have $\mathrm{R}^{\mathrm{LDF}}(T) \leq MT$. When $\frac{T}{\log T} \leq \frac{N}{M\theta^2}$, we therefore have $\mathrm{R}^{\mathrm{LDF}}(T) \leq \frac{N}{\theta^2}\log T$. We proceed now assuming $\frac{T}{\log T} \geq \frac{N}{M\theta^2}$.

Adding $\eta\sum_{i \in \mathcal{N}} \mu_i(d_i^*(t) - d_i(t))$ to both sides of (8) and taking expectations gives

$$\begin{aligned} &\mathbf{E}\left[\Delta V(t) + \eta\sum_{i \in \mathcal{N}} \mu_i(d_i^*(t) - d_i(t))\right] \\ &\leq \frac{N}{2} + \sum_{i \in \mathcal{N}} \mathbf{E}[Q_i(t)r_i - Q_i(t)d_i(t) + \eta\mu_i d_i^*(t) - \eta\mu_i d_i(t)] \\ &\overset{(a)}{\leq} \frac{N}{2} + \sum_{i \in \mathcal{N}} \mathbf{E}[Q_i(t)d_i^*(t) - Q_i(t)d_i(t) + \eta\mu_i d_i^*(t) - \eta\mu_i d_i(t)] \\ &= \frac{N}{2}\sum_{i \in \mathcal{N}} \mathbf{E}[d_i^*(t)(Q_i(t) + \eta\mu_i) - d_i(t)(Q_i(t) + \eta\mu_i)] \end{aligned} \tag{21}$$

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where (a) is because $\mathcal{S}^*(t)$ and $\mathbf{Q}(t)$ are independent and $r_i \leq \mathbf{E}[d_i^*(t)]$ according to (2). Now consider a super-arm $\mathcal{S}'(t) \in \operatorname{argmax}_{\mathcal{S}\in\mathbb{S}_{\mathcal{A}(t)}} \sum_{i\in\mathcal{S}}(Q_i(t)+\eta\mu_i)$. Then we have

$$\begin{aligned}
&\sum_{i\in\mathcal{N}} \mathbf{E}[d_i^*(t)(Q_i(t)+\eta\mu_i) - d_i(t)(Q_i(t)+\eta\mu_i)] \\
&= \mathbf{E}\left[\sum_{i\in\mathcal{S}^*(t)}(Q_i(t)+\eta\mu_i) - \sum_{i\in\mathcal{S}(t)}(Q_i(t)+\eta\mu_i)\right] \\
&\leq \mathbf{E}\left[\sum_{i\in\mathcal{S}'(t)}(Q_i(t)+\eta\mu_i) - \sum_{i\in\mathcal{S}(t)}(Q_i(t)+\eta\mu_i)\right] \\
&\overset{(a)}{\leq} \mathbf{E}\left[\sum_{i\in\mathcal{S}'(t)}(Q_i(t)+\eta\mu_i) - \sum_{i\in\mathcal{S}(t)}(Q_i(t)+\eta\mu_i)\right] \\
&\quad + \mathbf{E}\left[\sum_{i\in\mathcal{S}(t)}(Q_i(t)+\eta U_i(t)) - \sum_{i\in\mathcal{S}'(t)}(Q_i(t)+\eta U_i(t))\right] \\
&= \eta\,\mathbf{E}\left[\sum_{i\in\mathcal{S}'(t)}(\mu_i - U_i(t)) + \sum_{i\in\mathcal{S}(t)}(U_i(t)-\mu_i)\right]
\end{aligned} \tag{22}$$

where in (a), we are adding a positive quantity due to the definition of the LDF algorithm (4). Then using this bound in (21), summing over rounds $t \in \{1, 2, \ldots, T\}$ and dividing both sides by η gives

$$\begin{aligned}
&\frac{1}{\eta}\sum_{t=1}^{T}\mathbf{E}[\Delta V(t)] + \mathrm{R}^{\mathrm{LDF}}(T) \\
&\leq \frac{NT}{2\eta} + \sum_{t=1}^{T}\mathbf{E}\left[\sum_{i\in\mathcal{S}'(t)}(\mu_i - U_i(t)) + \sum_{i\in\mathcal{S}(t)}(U_i(t)-\mu_i)\right].
\end{aligned} \tag{23}$$

Define $C_1 \triangleq \sum_{t=1}^{T}\mathbf{E}\Big[\sum_{i\in\mathcal{S}'(t)}(\mu_i - U_i(t))\Big]$ and $C_2 \triangleq \sum_{t=1}^{T}\mathbf{E}\Big[\sum_{i\in\mathcal{S}(t)}(U_i(t)-\mu_i)\Big]$. Then we have

$$\mathrm{R}^{\mathrm{LDF}}(T) \overset{(a)}{\leq} \frac{1}{\eta}\sum_{t=1}^{T}\mathbf{E}[\Delta V(t)] + \mathrm{R}^{\mathrm{LDF}}(T) \overset{(b)}{\leq} \frac{NT}{2\eta} + C_1 + C_2. \tag{24}$$

where (a) is because $\sum_{t=1}^{T}\mathbf{E}[\Delta V(t)]$ is a nonnegative telescoping series and (b) is from (23). We bound C_1 by

$$C_1 \leq \sum_{t=1}^{T}\sum_{i\in\mathcal{N}}\mathbf{E}\big[[\mu_i - U_i(t)]^+\big] \overset{(a)}{\leq} \sum_{t=1}^{T}\sum_{i\in\mathcal{N}}\mathbf{E}\big[\mathbb{1}_{\{\mu_i > U_i(t)\}}\big] \tag{25}$$

where (a) is because $[\mu_i - U_i(t)]^+ = 0 = \mathbb{1}_{\{\mu_i > U_i(t)\}}$ in the event $\{\mu_i \leq U_i(t)\}$ and $[\mu_i - U_i(t)]^+ \leq 1 = \mathbb{1}_{\{\mu_i > U_i(t)\}}$ in the event $\{\mu_i > U_i(t)\}$. Let $X_{i,v}$ denote the v-th reward received from arm i. Then each of the quantities $\mathbf{E}\big[\mathbb{1}_{\{\mu_i > U_i(t)\}}\big]$ can be bounded by

$$\begin{aligned}
&\mathbf{E}\big[\mathbb{1}_{\{\mu_i > U_i(t)\}}\big] = \mathrm{P}(\mu_i > U_i(t)) \\
&= \mathrm{P}(\mu_i > U_i(t)\,,\, s_i(t-1) > 0) \\
&\quad + \mathrm{P}(\mu_i > U_i(t)\,,\, s_i(t-1) = 0) \\
&\overset{(a)}{=} \mathrm{P}(\mu_i > U_i(t)\,,\, s_i(t-1) > 0) \\
&\overset{(b)}{=} \mathrm{P}\left(\mu_i > \frac{\sum_{v=1}^{s_i(t-1)} X_{i,v}}{s_i(t-1)} + \sqrt{\frac{3\log t}{2s_i(t-1)}}\,,\, s_i(t-1) > 0\right) \\
&\overset{(c)}{\leq} \mathrm{P}\left(\bigcup_{m=1}^{t-1}\left\{\mu_i > \frac{\sum_{v=1}^{m} X_{i,v}}{m} + \sqrt{\frac{3\log t}{2m}}\right\}, s_i(t-1) > 0\right) \\
&\leq \mathrm{P}\left(\bigcup_{m=1}^{t-1}\left\{\mu_i > \frac{\sum_{v=1}^{m} X_{i,v}}{m} + \sqrt{\frac{3\log t}{2m}}\right\}\right) \\
&\overset{(d)}{\leq} \sum_{m=1}^{t-1}\mathrm{P}\left(\mu_i > \frac{\sum_{v=1}^{m} X_{i,v}}{m} + \sqrt{\frac{3\log t}{2m}}\right) \overset{(e)}{\leq} \sum_{m=1}^{t-1}\frac{1}{t^3} \leq \frac{1}{t^2}
\end{aligned} \tag{26}$$

where (a) is because we initialize $U_i(t) = 1$ when $s_i(t-1) = 0$, so $\mathrm{P}(\mu_i > U_i(t)\,,\, s_i(t-1) = 0) = 0$, in (b) we ignore the $\min(\,\cdot\,, 1)$ operation when rewriting $U_i(t)$ because it has no effect on the probability since $\mu_i \leq 1$, (c) is because $s_i(t-1) \in \{1, \ldots, t-1\}$ in the event $s_i(t-1) > 0$, in (d) we take the union bound and in (e) we apply Hoeffding's inequality. Using this bound in (25) gives

$$C_1 \leq \sum_{t=1}^{T}\sum_{i\in\mathcal{N}}\frac{1}{t^2} \leq N\sum_{t=1}^{\infty}\frac{1}{t^2} = \frac{N\pi^2}{6}. \tag{27}$$

Next we focus on bounding C_2. For each arm i, define the event $\mathcal{D}_i \triangleq \bigcap_{t=1}^{T}\left\{\delta_i(t) \leq \frac{2\log T}{\theta_0}\right\}$ where $\theta_0 \triangleq \sqrt{\frac{N\log T}{MT}}$. Then

$$\begin{aligned}
C_2 &= \sum_{i\in\mathcal{N}}\mathbf{E}\left[\sum_{t=1}^{T} d_i(t)(U_i(t)-\mu_i)\right] \\
&= \sum_{i\in\mathcal{N}}\mathbf{E}\left[\left(\mathbb{1}_{\mathcal{D}_i^C} + \mathbb{1}_{\mathcal{D}_i}\right)\sum_{t=1}^{T} d_i(t)(U_i(t)-\mu_i)\right] \\
&\leq \underbrace{T\sum_{i\in\mathcal{N}}\mathbf{E}\Big[\mathbb{1}_{\mathcal{D}_i^C}\Big]}_{\triangleq C_3} + \underbrace{\sum_{i\in\mathcal{N}}\mathbf{E}\left[\mathbb{1}_{\mathcal{D}_i}\sum_{t=1}^{T} d_i(t)(U_i(t)-\mu_i)\right]}_{\triangleq C_4}.
\end{aligned} \tag{28}$$

We proceed to bound C_3. Notice that each

$$\begin{aligned}
&\mathbf{E}\Big[\mathbb{1}_{\mathcal{D}_i^C}\Big] = \mathrm{P}\big(\mathcal{D}_i^C\big) \overset{(a)}{\leq} \sum_{t=1}^{T}\mathrm{P}\left(\delta_i(t) > \frac{2\log T}{\theta_0}\right) \\
&\overset{(b)}{\leq} \frac{1}{T^2}\sum_{t=1}^{T}\mathbf{E}\Big[e^{\theta_0\,\delta_i(t)}\Big] \overset{(c)}{\leq} \frac{1}{T^2}\sum_{t=1}^{T}\mathbf{E}\Big[e^{\theta\,\delta_i(t)}\Big] \leq \frac{D}{T}
\end{aligned} \tag{29}$$

where (a) is by the union bound, (b) is by the Chernoff bound and (c) is because by assumption, $T/\log T \geq N/(M\theta^2)$, so $\theta_0 \leq \theta$. Then $C_3 \leq ND$ and from (24), (27), and (28), we now have

$$\mathrm{R}^{\mathrm{LDF}}(T) \leq \frac{NT}{2\eta} + \frac{N\pi^2}{6} + ND + C_4. \tag{30}$$

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Focusing now on C_4, let $t_{i,a}$ denote the round in which arm i is played for the a-th time, so $d_i(t_{i,a}) = 1$ for all $a \in \{1, 2, \ldots, n_i(T)\}$ and $d_i(t) = 0$ for all other rounds t. Then we can rewrite C_4 in terms of the times each arm is played, i.e.

$$\begin{aligned} C_4 &= \sum_{i\in\mathcal{N}} \mathbf{E}\left[\mathbb{1}_{\mathcal{D}_i} \sum_{a=1}^{n_i(T)} (U_i(t_{i,a}) - \mu_i)\right] \\ &= \sum_{i\in\mathcal{N}} \mathbf{E}\left[\mathbb{1}_{\mathcal{D}_i} \sum_{a=1}^{\lfloor 2\log T/\theta_0\rfloor+1} (U_i(t_{i,a}) - \mu_i)\right] \\ &\quad + \sum_{i\in\mathcal{N}} \mathbf{E}\left[\mathbb{1}_{\mathcal{D}_i} \sum_{a=\lfloor 2\log T/\theta_0\rfloor+2}^{n_i(T)} (U_i(t_{i,a}) - \mu_i)\right] \\ &\overset{(a)}{\leq} 2\sqrt{NMT\log T} + N \\ &\quad + \sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=\lfloor 2\log T/\theta_0\rfloor+2}^{n_i(T)} \mathbb{1}_{\mathcal{D}_i}(U_i(t_{i,a}) - \mu_i)\right] \end{aligned} \tag{31}$$

where in (a), we bound $(U_i(t_{i,a}) - \mu_i) \leq 1$ for all $i \in \mathcal{N}$ and $a \in \{1, 2, \ldots, \lfloor 2\log T/\theta_0\rfloor + 1\}$. Next, we define the events

$$\mathcal{E}_i(t) \triangleq \left\{\hat{\mu}_i(t-1) - \mu_i \leq \sqrt{\frac{3\log t}{2s_i(t-1)}}\right\} \tag{32}$$

for all arms i and rounds t wherein $s_i(t-1) > 0$. Notice that $s_i(t-1) > 0$ for all $t > t_{i,\lfloor 2\log T/\theta_0\rfloor+1}$ in the event $\mathcal{D}_i$ because the event bounds all the delays by $2\log T/\theta_0$, so at least one reward will have been received $\lfloor 2\log T/\theta_0\rfloor$ rounds after the first time arm i was played. Then we can use this event to bound each $\mathbb{1}_{\mathcal{D}_i}(U_i(t_{i,a}) - \mu_i)$ for all $i \in \mathcal{N}$ and $a \in \{\lfloor 2\log T/\theta_0\rfloor + 2, \ldots, n_i(T)\}$ by

$$\begin{aligned} &\mathbb{1}_{\mathcal{D}_i}(U_i(t_{i,a}) - \mu_i) \\ &= \mathbb{1}_{\mathcal{D}_i}(U_i(t_{i,a}) - \mu_i)\left(\mathbb{1}_{\mathcal{E}_i^C(t_{i,a})} + \mathbb{1}_{\mathcal{E}_i(t_{i,a})}\right) \\ &\leq \mathbb{1}_{\mathcal{D}_i}\left(\mathbb{1}_{\mathcal{E}_i^C(t_{i,a})} + \mathbb{1}_{\mathcal{E}_i(t_{i,a})}(U_i(t_{i,a}) - \mu_i)\right) \\ &\overset{(a)}{\leq} \mathbb{1}_{\mathcal{D}_i}\left(\mathbb{1}_{\mathcal{E}_i^C(t_{i,a})} + \mathbb{1}_{\mathcal{E}_i(t_{i,a})}\left(2\sqrt{\frac{3\log t_{i,a}}{2s_i(t_{i,a}-1)}}\right)\right) \\ &\leq \mathbb{1}_{\mathcal{D}_i,\mathcal{E}_i^C(t_{i,a})} + \sqrt{\frac{6\log T\,\mathbb{1}_{\mathcal{D}_i}}{s_i(t_{i,a}-1)}} \end{aligned} \tag{33}$$

where (a) is by definition of the UCB term (6) and the event $\mathcal{E}_i(t)$ (32). Then using (33) to bound (31) gives

$$\begin{aligned} C_4 &\leq 2\sqrt{NMT\log T} + N \\ &\quad + \sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=\left\lfloor\frac{2\log T}{\theta_0}\right\rfloor+2}^{n_i(T)} \left(\mathbb{1}_{\mathcal{D}_i,\mathcal{E}_i^C(t_{i,a})} + \sqrt{\frac{6\log T\,\mathbb{1}_{\mathcal{D}_i}}{s_i(t_{i,a}-1)}}\right)\right] \end{aligned} \tag{34}$$

Next notice that for all for all $i \in \mathcal{N}$ and $a \in \{\lfloor 2\log T/\theta_0\rfloor + 2, \ldots, n_i(T)\}$, we have

$$\begin{aligned} &\mathbf{E}\left[\mathbb{1}_{\mathcal{D}_i}\mathbb{1}_{\mathcal{E}_i^C(t_{i,a})}\right] = \mathbf{P}\big(\mathcal{D}_i, \mathcal{E}_i^C(t_{i,a})\big) \\ &= \mathbf{P}\left(\mathcal{D}_i, \frac{\sum_{v=1}^{s_i(t_{i,a}-1)} X_{i,v}}{s_i(t_{i,a}-1)} - \mu_i > \sqrt{\frac{3\log t_{i,a}}{2s_i(t_{i,a}-1)}}\right) \\ &\overset{(a)}{\leq} \mathbf{P}\left(\mathcal{D}_i, \frac{\sum_{v=1}^{s_i(t_{i,a}-1)} X_{i,v}}{s_i(t_{i,a}-1)} - \mu_i > \sqrt{\frac{3\log a}{2s_i(t_{i,a}-1)}}\right) \\ &\overset{(b)}{\leq} \mathbf{P}\left(\mathcal{D}_i, \bigcup_{m=1}^{a-1}\left\{\frac{\sum_{v=1}^{m} X_{i,v}}{m} - \mu_i > \sqrt{\frac{3\log a}{2m}}\right\}\right) \\ &\leq \mathbf{P}\left(\bigcup_{m=1}^{a-1}\left\{\frac{\sum_{v=1}^{m} X_{i,v}}{m} - \mu_i > \sqrt{\frac{3\log a}{2m}}\right\}\right) \overset{(c)}{\leq} \frac{1}{a^2} \end{aligned} \tag{35}$$

where (a) is because $a \leq t_{i,a}$, (b) is because $s_i(t_{i,a}-1) \in \{1, \ldots, a-1\}$ in the event $\mathcal{D}_i$, and (c) is from the union bound followed by Hoeffding's inequality. Then

$$\sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=\lfloor 2\log T/\theta_0\rfloor+2}^{n_i(T)} \mathbb{1}_{\mathcal{D}_i,\mathcal{E}_i^C(t_{i,a})}\right] \leq N\sum_{a=1}^{\infty}\frac{1}{a^2} = \frac{N\pi^2}{6}. \tag{36}$$

We can further bound (34) using (36) by

$$\begin{aligned} C_4 &\leq 2\sqrt{NMT\log T} + N\left(1 + \frac{\pi^2}{6}\right) \\ &\quad + \underbrace{\sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=\lfloor 2\log T/\theta_0\rfloor+2}^{n_i(T)} \sqrt{\frac{6\log T\,\mathbb{1}_{\mathcal{D}_i}}{s_i(t_{i,a}-1)}}\right]}_{\triangleq C_5}. \end{aligned} \tag{37}$$

Recall that $s_i(t) = n_i(t) - g_i(t)$ for all arms i and rounds t and notice that $n_i(t_{i,a}-1) = a-1$ for all $a > 1$ and $g_i(t) \leq \lfloor 2\log T/\theta_0\rfloor$ for all t in the event $\mathcal{D}_i$, so

$$\begin{aligned} C_5 &\leq \sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=\lfloor 2\log T/\theta_0\rfloor+2}^{n_i(T)} \sqrt{\frac{6\log T\,\mathbb{1}_{\mathcal{D}_i}}{a-1-\lfloor 2\log T/\theta_0\rfloor}}\right] \\ &= \sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=1}^{n_i(T)-\lfloor 2\log T/\theta_0\rfloor-1} \sqrt{\frac{6\log T\,\mathbb{1}_{\mathcal{D}_i}}{a}}\right] \\ &\leq \sum_{i\in\mathcal{N}} \mathbf{E}\left[\sum_{a=1}^{n_i(T)} \sqrt{\frac{6\log T}{a}}\right] \overset{(a)}{\leq} \sqrt{6\log T}\,\mathbf{E}\left[\sum_{i\in\mathcal{N}} 2\sqrt{n_i(T)}\right] \\ &\overset{(b)}{\leq} \sqrt{6\log T}\,\mathbf{E}\left[2\sqrt{N\sum_{i\in\mathcal{N}} n_i(T)}\right] \leq \sqrt{24NMT\log T} \end{aligned} \tag{38}$$

where (a) is because each $\sum_{a=1}^{n_i(T)}\sqrt{1/a} \leq 1 + \int_{x=1}^{n_i(T)}\sqrt{1/x}\,\mathrm{d}x \leq 2\sqrt{n_i(T)}$ and (b) is by the Cauchy-Schwartz inequality. Then using (38) to bound (37) and (37) to bound (30) finishes the proof. □

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